

Numerical Assignment-01

1. What are the Truncation and round off errors in numerical analysis? Explain with an example?

Ans:

In numerical analysis:

Truncation error

Truncation errors are the difference between the actual value of the function and the truncated value of the given function. The truncated value of the functions is the approximated value up to a given number of digits.

Simple Example

Take the series for e^x (which is a way to calculate the value of e raised to x):

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

If we want to find e^1 (which is about 2.718), but we stop after the x^2 term:

$$e^1 \approx 1 + 1 + \frac{1^2}{2} = 2.5$$

The real value of e is about 2.718, so the truncation error (the difference between the exact and the approximate value) is:

$$\text{Error} = 2.718 - 2.5 \approx 0.218$$

Round-off Errors

Round-off errors occur because computers have limitations in how they store numbers. They can't keep every single digit of a long number, so they have to round or cut off some digits. This means that when we do calculations, we might not get the exact answer.

Example:

Take the square root of 2, which is about 1.41421356237. If a computer can only show three decimal places, it might display it as 1.414. By rounding the number, the computer loses some precision, which creates a round-off error. This small difference can add up, especially in calculations that involve many steps.

2. Discuss the necessity of the root of an equation. What are the characteristics of a root?**Ans:**

A **root** of an equation is a number that makes the equation true when plugged in. For example, in the equation $x^2 - 4 = 0$, the roots are 2 and -2—because both make the equation correct.

Why Are Roots Important?**1. Solving Problems:**

- Roots help us find solutions to equations quickly. They tell us which values work.

2. Understanding Functions:

- Roots show where a function crosses the x-axis. This helps us see how the function behaves.

3. Real-World Uses:

- In everyday life, roots can represent important points, like when a ball hits the ground or when a business breaks even.

4. Optimization:

- Roots help find maximum or minimum values in problems, like maximizing profit or minimizing costs.

Characteristics of a root:

1. **Satisfies the Equation:** Substituting the root into the equation results in a true statement.
2. **Intersection with Axes:** For functions, a root corresponds to the x-value where the graph intersects the x-axis.
3. **Multiplicity:** A root can have multiplicity, indicating how many times it repeats (e.g., a double root touches the x-axis but doesn't cross it).
4. **Complexity:** Roots can be real or complex numbers.
5. **Dependence on Coefficients:** The nature and number of roots depend on the coefficients and degree of the polynomial.

3. What are the popular methods of finding a root? Discuss the differences between opening and bracketing methods. What are the advantages and disadvantages of the corresponding methods?

Ans:

Popular Methods of Finding a Root:

1. **Bisection Method**
2. **Newton-Raphson Method**
3. **Secant Method**
4. **False Position Method (Regula Falsi)**
5. **Fixed-Point Iteration**

Differences between Bracketing and Open Methods:

Bracketing Methods

Definition:

Bracketing methods are techniques for finding roots of equations that start with two initial guesses. These guesses help us know that the root lies between them.

Examples:

- **Bisection Method:** This method repeatedly divides the interval in half. If the root is between the two guesses, the method narrows down the interval until it finds the root.
- **False Position Method:** The **False Position Method** (also called the **Regula Falsi Method**) is a numerical technique used to find the roots of

a function. It is based on the principle of linear interpolation. The method starts with two initial guesses x_0 and x_1 such that the function values at these points have opposite signs (i.e., $f(x_0) \cdot f(x_1) < 0$), which indicates that a root lies between them.

Open Methods

Definition:

Open methods are techniques that don't require the root to be bracketed. They can start with one or more guesses without needing to know that the root is between them.

Examples:

- **Newton-Raphson Method:** This method uses tangent lines to find the root. It starts with one guess and refines it using the derivative of the function.
- **Secant Method:** This method uses two initial guesses but does not require them to bracket the root. It approximates the derivative to find better guesses.

Here's a chart summarizing the differences between bracketing and open methods for finding roots:

Feature	Bracketing Methods	Open Methods
Initial Points	Requires two initial points (a, b)	Requires one or two initial points
Convergence Guarantee	Guaranteed if conditions are met	No guarantee of convergence
Speed of Convergence	Generally slower	Can be faster
Function Behavior	Works for any continuous function	Requires knowledge of function behavior
Derivatives Needed	No derivatives required	Derivatives required (e.g., Newton's method)
Complexity	Simpler to implement	More complex, especially for derivatives
Interval Dependency	Needs an interval where the function changes sign	Can work without a bracketing interval
Robustness	More robust for poorly behaved functions	Less robust, may fail for certain functions

Advantages and Disadvantages:

Bracketing Methods

Advantages:

1. **Guaranteed Convergence:** Always converges if the initial guesses bracket the root.
2. **Simplicity:** Easy to understand and implement, making them accessible for beginners.
3. **Robustness:** Less sensitive to the choice of initial guesses; as long as the root is bracketed, convergence is assured.
4. **Well-Suited for Non-Differentiable Functions:** Can be applied even when the function is not differentiable in the interval.
5. **Stable:** Typically less affected by numerical errors, providing more reliable results.

Disadvantages:

1. **Slower Convergence:** Generally slower convergence rates compared to open methods.
2. **Function Evaluations:** They need calculations at both ends of the interval, which can make the process slower.
3. **Limited Precision:** May require many iterations to achieve high precision, making them less efficient for complex problems.
4. **Interval Management:** The need to continuously narrow the interval can complicate implementation.
5. **Not Suitable for Multiple Roots:** Can struggle with functions that have multiple roots within the bracketed interval.

Open Methods

Advantages:

1. **Faster Convergence** Generally faster convergence rates (quadratic for Newton-Raphson).
2. **No Bracketing Needed:** Do not require an initial interval to bracket the root, making them more flexible.
3. **Fewer Iterations:** Typically require fewer iterations, reducing overall computation time.
4. **Efficient for Smooth Functions:** Highly effective when the function is smooth and derivatives are available.
5. **Adaptable:** Can be adjusted or combined with other methods for improved performance on specific problems.

Disadvantages:

1. **Risk of Divergence:** Not guaranteed to converge; may diverge with poor initial guesses or for certain functions.
2. **Derivative Requirement:** Some methods (e.g., Newton-Raphson) require the computation of derivatives, which can be complex.
3. **Complex Implementation:** Can be more difficult to implement, especially for methods that require careful handling of derivatives.
4. **Sensitivity to Initial Guesses:** Performance heavily depends on the quality of initial guesses; poor choices can lead to failure.
5. **Not Suitable for Non-Differentiable Functions:** Cannot be effectively used for functions that are not smooth or have discontinuities.

In summary, bracketing methods are more robust but slower, while open methods are faster but require better initial guesses and may not always converge.

4. Consider the equation $y = 5\cos^2 x - e^{2x}$

Find the root of the above equation using false position and secant methods up to 5 iterations.

5. Can a bisection root finding method be divergent? Justify your answer with the help of an example.

Ans:

The Bisection method cannot diverge if applied correctly. It always converges to a root within a bracketing interval, provided the following conditions are met:

Justification

The Bisection method has the following characteristics:

- It requires two initial guesses, a and b , such that $f(a)$ and $f(b)$ have opposite signs.
- It repeatedly bisects the interval and selects the subinterval where a sign change occurs, ensuring that the root remains within the interval.

Conditions for Convergence

1. **Continuous Function:** The function must be continuous on the interval $[a, b]$.
2. **Sign Change:** There must be a sign change between $f(a)$ and $f(b)$ (i.e., $f(a) \cdot f(b) < 0$).

Example

Consider the function:

$$f(x) = x^2 - 1$$

Choose the interval $[0, 2]$:

- $f(0) = -1$ (negative)
- $f(2) = 3$ (positive)

Since $f(0) \cdot f(2) < 0$, the Bisection method will converge to the root $x=1$.

The **bisection method** cannot be divergent as long as the initial interval $[a,b]$ satisfies the condition that $f(a)$ and $f(b)$ have opposite signs, i.e., $f(a) \cdot f(b) < 0$. This ensures that a root exists between a and b , and the method will always converge to the root because the interval is repeatedly halved, reducing the error with each iteration.

6. What is the slope of an equation? What is the significance of slope for Newton's method of root finding? Discuss with an example.

Ans:

The **slope** refers to the steepness or rate of change of a line or curve. For a straight line, it's the ratio of the vertical change (rise) to the horizontal change (run), given by:

$$slope = \frac{\Delta y}{\Delta x}$$

For a function $f(x)$, the slope at a point $x=a$ is the derivative $f'(a)$, which describes how the function changes at that point.

Significance of Slope in Newton's Method

- ✚ **Direction of Update:** The slope indicates how to adjust the current guess. A steeper slope leads to larger steps towards the root.
- ✚ **Magnitude of Update:** The formula $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ shows that the update size depends on the slope. A larger slope allows for quicker adjustments.
- ✚ **Speed of Convergence:** When you're close to the root, the method gets much faster (this is called quadratic convergence). If the slope is small, it can slow you down or even make you stuck.
- ✚ **Behavior Prediction:** Analyzing the slope helps understand how the function behaves near the root, affecting the number of iterations needed.

Significance of Slope in Newton's Method

In Newton's Method, the slope helps find a better approximation of the root using:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

- **Convergence:** A steeper slope (larger $|f'(x_n)|$) leads to faster convergence.
- **Direction:** The slope's sign determines the direction of the next approximation.

Example

For the function:

$$f(x) = x^2 - 2$$

1. **Initial Guess:** $x_0 = 1$
2. **Calculate Slope:**

$$f'(x) = 2x \quad \Rightarrow \quad f'(1) = 2$$

3. **Newton's Method Update:**

$$x_1 = 1 - \frac{f(1)}{f'(1)} = 1 - \frac{-1}{2} = 1.5$$

4. **Next Iteration:**

- $f(1.5) = 0.25$, $f'(1.5) = 3$
- Update:

$$x_2 = 1.5 - \frac{0.25}{3} \approx 1.4167$$

Conclusion

The slope is crucial in Newton's Method for determining the next approximation and ensuring convergence to the root efficiently.

7. Discuss the possible loop breaking conditions in numerical root finding methods. Demonstrate how they work .

Ans:

In numerical root-finding methods, loops break when specific conditions are met to ensure convergence or prevent infinite iterations. Key conditions include:

1. **Absolute Error:** The difference between successive approximations is small:

$$|x_{n+1} - x_n| < \epsilon$$

- **Example:** In Newton-Raphson, if changes in estimates become negligible, the loop stops.

2. **Relative Error:** The relative difference between approximations is small:

$$\frac{|x_{n+1} - x_n|}{|x_{n+1}|} < \epsilon$$

3. **Residual (Function Value) Tolerance:** The function value at the estimate is close to zero:

$$|f(x_{n+1})| < \epsilon$$

- **Example:** In the bisection method, if $|f(c)|$ is small, the loop stops.

4. **Maximum Iteration Limit:** The loop stops when the number of iterations exceeds a predefined limit:

$$n \geq N_{\max}$$

5. **Stagnation:** If the method stops making progress (i.e., no significant change in estimates), the loop breaks:

$$|x_{n+1} - x_n| = 0$$

Demonstration Example: Newton's Method with Tolerance

Consider solving $f(x) = x^2 - 2 = 0$ (the square root of 2) using Newton's method.

1. **Iteration formula:**

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} = x_n - \frac{x_n^2 - 2}{2x_n}$$

2. **Initial guess:** $x_0 = 1.5$, and set a tolerance $\epsilon = 10^{-6}$.

3. **Iterations:**

- Compute $f(1.5) = 1.5^2 - 2 = 0.25$, and $f'(1.5) = 2(1.5) = 3$.
- Use the Newton formula:

$$x_1 = 1.5 - \frac{0.25}{3} = 1.5 - 0.0833 = 1.4167$$

- Continue iterating, and stop when $|x_{n+1} - x_n| < 10^{-6}$.

Final Output:

When the difference between x_{n+1} and x_n becomes smaller than 10^{-6} , the loop breaks, and the root is approximated as $\sqrt{2} \approx 1.414213$.

8. Discuss the Trapezoidal rule of integration. Identify the significance of error in the Trapezoidal rule.

The **Trapezoidal Rule** is a numerical method used to approximate the definite integral of a function. It works by dividing the area under a curve into a series of trapezoids, calculating the area of each trapezoid, and then summing these areas to estimate the total integral.

Formula:

For a function $f(x)$ over an interval $[a, b]$, divided into n subintervals of equal width h , the Trapezoidal Rule is given by:

$$\int_a^b f(x)dx \approx h \left(\frac{y_0 + y_n}{2} + y_1 + y_2 + \cdots + y_{n-1} \right)$$

where:

- $h = \frac{b-a}{n}$ is the width of each subinterval,
- $y_i = f(x_i)$ are the function values at points x_i (including the endpoints $x_0 = a$ and $x_n = b$),
- The term $\frac{y_0 + y_n}{2}$ represents the average of the function values at the endpoints $y_0 = f(a)$ and $y_n = f(b)$.

Intuition:

- The term $\frac{y_0 + y_n}{2}$ accounts for the contributions from the two endpoints of the interval.
- The sum $y_1 + y_2 + \cdots + y_{n-1}$ adds the function values at the intermediate points.
- The total sum is multiplied by h , representing the width of each subinterval, giving an approximation of the total integral.

Significance of error in the Trapezoidal rule.

The significance of the error in the **Trapezoidal Rule**, based on the error formula:

$$E_T = -\frac{(b-a)^3}{12n^2} f''(\xi)$$

is as follows:

1. **Accuracy of Approximation:** The error tells how close the trapezoidal approximation is to the true value of the integral. Smaller error means better accuracy.
2. **Effect of Curvature:** The second derivative $f''(\xi)$ in the formula shows that if the function is highly curved, the error is larger. For linear functions, the error is zero.
3. **Impact of Subintervals (n):** The error decreases as the number of subintervals n increases. More intervals lead to a better fit of the trapezoids to the curve, improving the approximation.
4. **Balance Between Cost and Precision:** Understanding the error helps to balance between computational effort (more subintervals) and the level of precision needed.

9. Identify the area intersected by the curve $f(x) = 5x^4 - 3x^3 + 6x^2 - 4$ between $x=0$ to 5, using the Simpson $\frac{3}{8}$ rule.

Ans:

To find the area under the curve

$f(x)=5x^4-3x^3+6x^2-4$ between $x=0$ and $x=5$ using the Simpson $\frac{3}{8}$ Rule, we follow these steps:

Simpson $\frac{3}{8}$ Rule Formula

The Simpson $\frac{3}{8}$ Rule is given by:

$$\int_a^b f(x) dx \approx \frac{3h}{8} (f(x_0) + 3f(x_1) + 3f(x_2) + 2f(x_3) + \dots + f(x_n))$$

where $h = \frac{b-a}{n}$ and n must be a multiple of 3.

Step 1: Define the Interval

- $a = 0, b = 5$

Step 2: Choose n

- Let $n = 3$ (multiple of 3).

Step 3: Calculate h

$$h = \frac{b - a}{n} = \frac{5 - 0}{3} = \frac{5}{3}$$

Step 4: Calculate x_i

- $x_0 = 0$
- $x_1 = \frac{5}{3}$
- $x_2 = \frac{10}{3}$
- $x_3 = 5$

Step 5: Evaluate $f(x_i)$

1. $f(0) = -4$
2. $f\left(\frac{5}{3}\right) \approx 3.74$
3. $f\left(\frac{10}{3}\right) \approx 172.22$
4. $f(5) = 2900$

Step 6: Apply Simpson $\frac{3}{8}$ Rule

$$\int_0^5 f(x) dx \approx \frac{3h}{8} \left(f(0) + 3f\left(\frac{5}{3}\right) + 3f\left(\frac{10}{3}\right) + f(5) \right)$$

Substituting the values:

$$= \frac{3 \cdot \frac{5}{3}}{8} (-4 + 3(3.74) + 3(172.22) + 2900)$$

$$= \frac{5}{8} (-4 + 11.22 + 516.66 + 2900)$$

Combining:

$$= \frac{5}{8} (3423.88) \approx 2140.86$$

Conclusion

The area under the curve from $x = 0$ to $x = 5$ is approximately **2140.86**.

10. Discuss the difference between interpolation and approximation with example.

Ans:

Interpolation vs. Approximation**Interpolation:**

- **Definition:** Interpolation is the method of estimating unknown values that lie between known data points. It helps us predict values in a smooth way based on nearby data. This approach is useful in many areas, like math and engineering, where we need to fill in gaps or create a complete picture from limited information.
- **Example:** Given points (1,2) (2,3) and (3,5) using linear interpolation to estimate the value at $x=2.5$ would give a value that lies exactly on the line connecting (2,3) and (3,5).

Approximation:

- **Definition:** Approximation is a method of estimating values or functions that are **not exactly known**. It typically uses simpler mathematical models to represent data trends. The goal is to capture the overall behavior rather than fit every data point precisely. This approach is valuable in fields like physics, engineering, and statistics. It allows for easier calculations and insights into complex problems.
- **Example:** If you want to estimate the square root of 10, you might approximate it as 3.16, knowing that the exact value is about 3.162. This is useful when exact precision isn't required

Interpolation and approximation are both methods used to estimate unknown values of a function, but they differ in their purpose and how they treat data.

Interpolation

- **Definition:** Interpolation is the process of estimating the value of a function at a point within the range of known data points by fitting a curve (or line) through those points.
- **Goal:** To find an exact value that passes through the given data points.
- **Key Feature:** The interpolating function matches the known data points exactly.

Example: Suppose we have the following data points:

$$(1, 2), (2, 4), (3, 6)$$

If we want to interpolate the value at $x = 2.5$, we could use linear interpolation between $(2, 4)$ and $(3, 6)$, which would give $f(2.5) = 5$. This interpolated value lies exactly on the line formed by the known data points.

Approximation

- **Definition:** Approximation involves estimating a function or value by constructing a model that does not necessarily pass through all data points but provides a close estimate of the overall trend.
- **Goal:** To obtain a function that is close to the actual data but not necessarily exact at any given point.
- **Key Feature:** The approximating function minimizes the error between the estimated values and the actual data points.

Example: For the same set of points $(1, 2), (2, 4), (3, 6)$, an approximating function could be $f(x) = 2x$, a simple straight line that fits the trend of the data. If you introduce another point, say $(4, 7.8)$, this line may not pass exactly through the point $(4, 7.8)$, but it still gives a good estimate of the overall trend.